



### Lab 1: Ohm's Law

#### Objectives

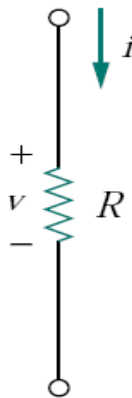
- Find the resistance of a resistor from its color code.
- Measure voltage, current and resistance values using a digital multimeter.
- Verify the validity of Ohm's Law.

#### Verification of Ohm's Law

**Theory:** The relationship between current and voltage for a resistor is known as **Ohm's law**.

**Ohm's law** states that the current through a conductor between two points is directly proportional to the voltage across the two points.

Or, the voltage across a resistor is directly proportional to the current flowing through the resistor.



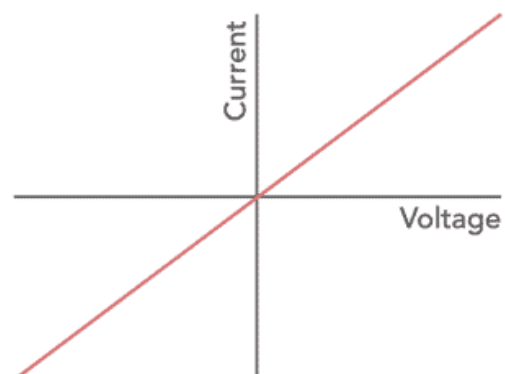
$$V \propto I \quad (1)$$

Ohm defined the constant of proportionality for a resistor to be the resistance,  $R$ . The resistance  $R$  of an element denotes its ability to resist the flow of electric current; it is measured in ohms ( $\Omega$ ).

Now the equation (1) becomes

$$\rightarrow V = IR$$

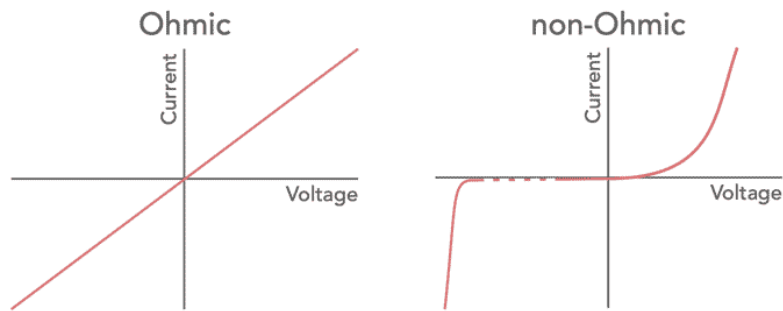
$$\rightarrow I = \frac{V}{R}$$



Ohmic conductors obey Ohm's Law, meaning their resistance remains constant regardless of voltage or current changes. Common examples include metals like copper and aluminum.

Non-Ohmic conductors, such as diodes, LED, and filament lamps, have resistance that varies with applied voltage or current. This is why a filament lamp’s brightness increases non-linearly as voltage rises.

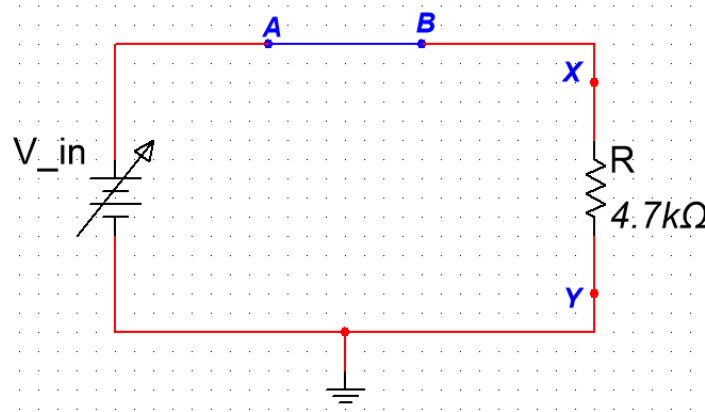
The resistance of a material is influenced by temperature—metals typically increase in resistance when heated, while semiconductors may decrease in resistance.



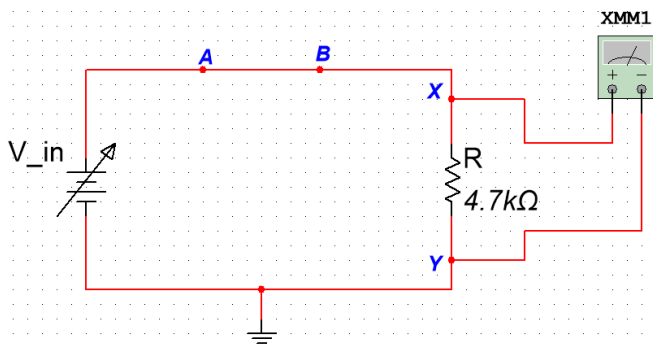
**List of Components:**

| Component Name             | Value (or Model Number) | Quantity |
|----------------------------|-------------------------|----------|
| Breadboard                 |                         | 1        |
| DC Power Supply (Variable) |                         | 1        |
| Resistors                  | 4.7 k $\Omega$          | 1        |
|                            | 10 k $\Omega$           | 1        |
| Connecting Wires           |                         | 10       |
| Digital Multimeter (DMM)   |                         | 1        |

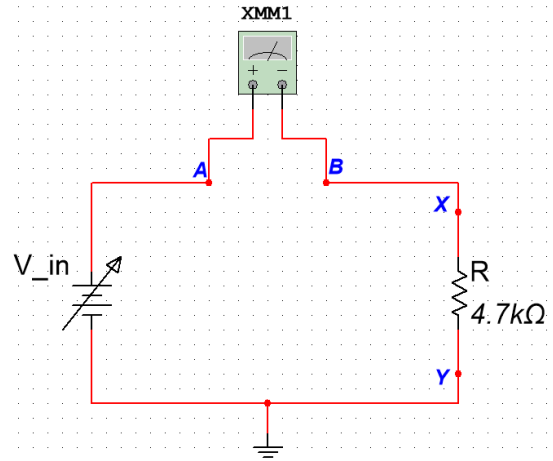
## Circuit Diagram:



Circuit for Experiment



Voltage Measurement: DMM in Parallel



Current Measurement: DMM in Series

## Procedure:

1. Identify the given resistors using color coding and fill in the required columns in Table 1.
2. Measure the resistances of the resistors using the DMM and fill in the required column in Table 1.
3. Calculate the percentage error of the resistance values.

$$\text{Percentage of Error} = \frac{|(\text{Practical value} - \text{Theoretical value})|}{\text{Theoretical value}} \times 100\%$$

4. Build circuit using the 4.7 kΩ resistor.
5. Set the voltage source to 2 V.
6. Connect the DMM (Voltmeter Mode) in parallel to the resistor (point X and Y).
7. Check the voltage across the resistor. Note it down in Table 2(a)
8. Increase the voltage and fill up Table 2(a)
9. Set the voltage source to 2 V. Disconnect the wire between source and resistor (point A and B). Connect the DMM (Ammeter Mode) there.
10. Measure the current flowing through the resistor. Note it down in Table 2(b).
11. Increase the voltage and fill up Table 2(b)
12. Calculate the power using the experimental values of I and R (Power =  $I^2R$ ).
13. Repeat step 4-12 for the 10 kΩ resistor. Record data in Table 3(a) and 3(b).

## Data Collection:

### Lab 1:

Group No. \_\_\_\_\_

Instructor's Signature \_\_\_\_\_

Table 1:

|                | Resistance using colour coding |        |        |        |                      | Resistance using DMM | % of Error |
|----------------|--------------------------------|--------|--------|--------|----------------------|----------------------|------------|
|                | Band 1                         | Band 2 | Band 3 | Band 4 | Resistance $\pm$ tol |                      |            |
| 4.7 k $\Omega$ |                                |        |        |        |                      |                      |            |
| 10 k $\Omega$  |                                |        |        |        |                      |                      |            |

Table 2: Experimental readings for 4.7 k $\Omega$

|                 | Voltage Measurement | Current Measurement | Power Calculation |
|-----------------|---------------------|---------------------|-------------------|
| Supply $V_{in}$ | Voltage, $V_R$ (V)  | Current, I (mA)     | Power, VI (mW)    |
| 2V              |                     |                     |                   |
| 4V              |                     |                     |                   |
| 6V              |                     |                     |                   |
| 8V              |                     |                     |                   |
| 10V             |                     |                     |                   |

Table 3: Experimental readings for 10 k $\Omega$

|                 | Voltage Measurement | Current Measurement | Power Calculation |
|-----------------|---------------------|---------------------|-------------------|
| Supply $V_{in}$ | Voltage, $V_R$ (V)  | Current, I (mA)     | Power, VI (mW)    |
| 2V              |                     |                     |                   |
| 4V              |                     |                     |                   |
| 6V              |                     |                     |                   |
| 8V              |                     |                     |                   |
| 10V             |                     |                     |                   |

*"If the circuit works, means you've just mastered a building block of modern tech"*

### Questions

1. Plot V (voltage) vs. I (current) graph for each resistor value in same graph.
2. Does your experimental circuit follow ohm's law? Explain how you figured it out.
3. Calculate the resistance of each circuit using the slope of your V vs. I graph. Compare these R graph values to the measured R values using DMM. Find the percent difference.
4. Based on Ohm's law, describe the linear relationship between voltage and current.

## MultiSim Simulation

- Verify Ohm's Law using MultiSim. Use resistor =  $50\Omega$  and  $3.3k\Omega$